1. Problem Statement & Clinical Pain Points

Small-to-medium hospitals and polyclinics in India face severe operational and clinical friction:

- **Paper Slips & Fragmented Billing:** OPD consultation fees, pharmacy medications, and pathology tests are billed on disconnected paper pads or separate single-function software, resulting in unbilled services and revenue leakage.
- **Queue Chaos & Patient Anxiety:** Overcrowded waiting areas with shouting token numbers create patient frustration and front-desk conflict.
- **Medication Safety Blind Spots:** Doctors prescribing under extreme time constraints lack automated warnings for lethal Drug-Drug Interactions (DDIs) or critical physiological thresholds (e.g. Stage 2 hypertension, hypoxia).
- **Internet Vulnerability:** Pure cloud systems grind to a complete halt during frequent broadband fiber cuts or ISP disruptions.

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2. Cycle Appetite

- **Cycle Appetite: Fixed 4-Week Delivery Window (Weeks 1–4)**.

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3. Fat-Marker Solution Shape (The 4-Week Scope)

Week 1 — End-to-End OPD-to-IPD Core Flow & 6-Day IPD Book

Registration • Queue • OPD Chamber • IPD Bed Admission • Forms 8, 9, 10 • Daily Treatment Form 29 • Intake/Output Form 61 • Discharge • Bill



Week 2 — Deterministic CDSS & Clinical Heuristics

Relational DDI Rules • Patient Allergy Screening • Vital Sign Threshold Alerts (NEWS2) • Rule-Based Discharge Summary Assembler



Week 3 — Hospital Automation & Print Templates

Nightly Bed Rent Job • Token Rollover • Print Engine (Rx, Form 8/9, Discharge Bill, 80mm Token) • Edge Docker LAN Sentinel



Week 4 — Comprehensive Pytest, E2E Testing & Hardening

Full OPD-to-IPD Lifecycle Tests • Concurrency Locks • CDSS Safety Suites •

Financial Rupee Balance
Audits • Demo Sign-off

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4. Week-by-Week Milestone Roadmap

Week 1: Complete End-to-End Clinical Flow (OPD to IPD & 6-Day IPD Book Core)

- **Objective:** Deliver the complete, functional clinical vertical slice from Outpatient arrival to Inpatient discharge within Week 1, fully aligned with the hospital's client dossier (`IPD Book 6 Days.pdf`).
- **Key Deliverables:**
  - **Platform Foundations:** MySQL 8.0 schema on Contabo LAMP, Argon2id auth, JWT refresh token rotation, RBAC guards, and monotonic concurrency locks (`UHID`, `OPD`, `IPD`, `BILL`).
  - **OPD Flow:** Patient registration (`POST /api/v1/patients`), daily doctor queue token assignment (`OPD-001001`), nurse triage vitals, and doctor consultation chamber with prescription creation and "Admit to IPD" referral order.
  - **IPD Admission & Bed Assignment:** Support both Doctor OPD referral and Direct Emergency/Walk-in admission. Bed matrix allocation, MLC (Medico-Legal Case) tagging, Next-of-Kin/Guardian capture, advance deposit payment, and digital generation of **Admission Consent (AH/MRD/FORM - 8)** and **Admission Form (AH/MRD/FORM - 9)**.
  - **Inpatient Stay Records (Client 6-Day Book Integration):**
    - **Doctor H&P Admission Note (AH/MRD/FORM - 10):** History, physical examination, provisional diagnosis.
    - **Treatment Sheet & Medication Administration Record (AH/MRD/FORM - 29):** Doctor drug orders (dosage, route, frequency, start/stop times), O2 therapy parameters (nasal prongs/mask, L/min), and nurse administration/omission logging.
    - **Intake and Output Chart (AH/MRD/FORM - 61):** Shift-wise recording of intake (Oral/RT, IV fluids, Blood) and output (Urine, RT aspiration/vomitus, Stool, Surgical Drain) with running net fluid balance.
    - **Daily Vitals Signs Chart:** Temperature, Pulse, Resp, BP, SpO2, and Pain Score (0–10).
    - **Nurse Progress Notes (AH/MRD/FORM - 72):** Shift-by-shift nursing handover and patient observations.
  - **Inpatient Discharge & Settlement:** Doctor signed discharge summary (`status = 'SIGNED'`), bed release, and consolidated 10-head final invoice generation with balance reconciliation.

Week 2: Deterministic Clinical Decision Support (CDSS) & Heuristics Engine

- **Objective:** Layer deterministic, sub-5 millisecond clinical safety screening and heuristic automation over the working Week 1 flow (100% rule-based and auditable; strictly NO ambient audio, speech-to-text, or non-deterministic LLMs in this MVP).
- **Key Deliverables:**
  - **Deterministic CDSS Safety Engine (Module 9):** Real-time bilateral Drug-Drug Interaction (DDI) collision detection querying relational database rules (`cdss_ddi_rules`) with severity levels (`CONTRAINDICATED`, `MAJOR`, `MODERATE`). Mandatory written doctor override justifications logged immutably to `cdss_alerts_log`.
  - **Patient Allergy Cross-Screening:** Active ingredient cross-matching against patient known allergies (`allergies_json` on patient profile) when prescribing medications in OPD or IPD Treatment Sheets (Form 29).
  - **Heuristic Vital Signs & Early Warning Engine:** Automated heuristic alerting for critical physiologic thresholds (severe hypoxia SpO2 < 90%, hypertensive crisis BP > 180/120, pediatric tachycardia, high fever) based on standard clinical scorecards (NEWS2 / MEWS).
  - **Rule-Based Inpatient Discharge Summary Assembler:** Deterministic template assembler that programmatically extracts and formats the patient's multi-day stay records (synthesizing Admission H&P Form 10, Treatment MAR administration records Form 29, daily vital trends, and lab test results into structured discharge sections without hallucination or LLM dependency).
  - **Heuristic Queue Prioritization:** Automatic tagging of high-acuity/abnormal triage vitals to flag urgent patients for earlier consultation in the OPD queue.

Week 3: Background Automation, Print Templates & Edge LAN Fallback

- **Objective:** Implement automated hospital operations, pixel-perfect physical print outputs, and local network failover.
- **Key Deliverables:**
  - **Automated Hospital Background Jobs:**
    - Nightly Bed-Rent Billing Job (midnight accrual of room tariffs per active bed allocation).
    - Midnight OPD Queue Rollover & No-Show Auto-Closure.
    - Automated Daily Database Backup & S3/Contabo Cold Storage Upload.
  - **Pixel-Perfect Print Templates (D20 Integration):**
    - OPD Prescription Slip (A4 doctor header, Rx grid, dosage schedule, follow-up).
    - OPD 80mm Thermal Queue Token Slip with QR code.
    - IPD Admission Face Sheet & Consent Form (Forms 8 & 9).
    - IPD Consolidated Discharge Bill (10 standard NABH charge heads).
    - Diagnostic Lab Report with reference ranges and abnormal flags.
  - **Local Edge Docker Fallback:** `docker-compose.edge.yml` configuration with SQLite/MySQL edge replica, conflict-free sequence partitioning (`OFFLINE-UHID-`

`*`), and background reconciliation sentinel when WAN internet recovers.

Week 4: Comprehensive Automated Testing, Hardening & Release Gate

- **Objective:** Exhaustive automated test coverage, load testing, security audits, and client acceptance sign-off.
- **Key Deliverables:**
  - **Pytest Suite:** 100% test coverage across critical clinical and financial paths:
    - End-to-end OPD-to-IPD patient journey lifecycle tests.
    - Concurrency tests: race condition validation on bed allocations and sequence counters under 50 simultaneous threads.
    - CDSS safety verification: 100% interception of lethal test drug pairs (Warfarin + Aspirin, Methotrexate + NSAIDs).
    - Financial balance audit: exact rupee matching (`subtotal - discount + tax == paid + balance`) across all invoices.
  - **Security & NABH Audit Trail:** Comprehensive access control validation (RBAC matrix) and immutable audit log verification.
  - **Client Demo Sign-Off:** Full-scale demonstration of the operational hospital flow with real seed data matching the client's oncology and general surgical departments.

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5. Developer Work Breakdown (Backend vs. Frontend)

To maximize development velocity and minimize merge conflicts, work is divided between two developers only: one Backend Developer and one Frontend Developer. Both developers own their respective implementations, coordinate through agreed API contracts, and review each other's integration points.

Module / Component	Primary Owner	Collaborator	Key Responsibilities
Database DDL & Schema Updates	Backend Developer	Frontend Developer	IPD book tables, indexes, constraints, migration scripts, and data models
Auth & Security Framework	Backend Developer	Frontend Developer	Argon2id password hashing, JWT rotation, RBAC permission guards, secure session handling, and frontend route protection
System Sequences Engine	Backend Developer	Frontend Developer	Concurrency-safe monotonic numbering for UHID, OPD, and IPD; displaying generated identifiers
Patient Registration & MPI (M1)	Backend Developer	Frontend Developer	Patient CRUD APIs, Aadhaar salted hashing, emergency contacts, registration forms, and patient search

Module / Component	Primary Owner	Collaborator	Key Responsibilities
OPD Queue & Token Engine (M2)	Backend Developer	Frontend Developer	Daily token generation, doctor calling APIs, queue management, and SSE-powered TV display
Vitals & Triage (M3)	Backend Developer	Frontend Developer	Vitals APIs, BMI calculation, abnormal-range flags, and triage entry interfaces
Doctor Consultation (M4)	Backend Developer	Frontend Developer	SOAP draft/autosave APIs, ICD-10 linking, consultation UI, and admission referral workflow
IPD Admissions & Beds (M6)	Backend Developer	Frontend Developer	Bed allocation, Form 8 and 9 consent workflows, MLC flag, advance deposit, and bed management UI
IPD Treatment & MAR (Form 29)	Backend Developer	Frontend Developer	Medication orders, oxygen therapy, nurse administration logs, and medication administration UI
IPD Intake & Output (Form 61)	Backend Developer	Frontend Developer	Oral/IV fluid intake, urine/drain output, shift records, and fluid balance charts
IPD Discharge & 10-Head Bill	Backend Developer	Frontend Developer	Discharge APIs, summary-signing workflow, bed release, consolidated billing ledger, and bill presentation
Deterministic CDSS Engine (M9)	Backend Developer	Frontend Developer	Drug–drug interaction rules, allergy cross-checks, clinical warnings, override logging, and alert interfaces
Vital Alerting & Heuristics	Backend Developer	Frontend Developer	NEWS2/MEWS thresholds, pediatric tachycardia rules, hypoxia flags, and real-time alert display
Discharge Summary Assembler	Backend Developer	Frontend Developer	Deterministic assembly of Forms 10, 29, and 61 records; summary preview and print interface
Background Automation Jobs	Backend Developer	Frontend Developer	Nightly bed rent, token rollover, database backup scripts, job monitoring, and status display
Print Template Engines (D20)	Frontend Developer	Backend Developer	`@media print` CSS, thermal 80 mm and A4 layouts, printable clinical forms, and API data integration
Edge Docker LAN Fallback	Backend Developer	Frontend Developer	Docker Compose bundle, local deployment configuration, offline-sync sentinel, connectivity status, and fallback UI

Module / Component	Primary Owner	Collaborator	Key Responsibilities
Automated Test Suites	Backend Developer	Frontend Developer	Backend integration/concurrency/CDSS safety tests; frontend component, workflow, and end-to-end tests

Working agreement

- Backend Developer: Owns database design, migrations, business logic, APIs, authentication, security, clinical rules, background jobs, and backend tests.
- Frontend Developer: Owns application UI, forms, dashboards, queue displays, clinical workflows, accessibility, responsive layouts, printing, and frontend tests.
- Shared responsibility: API contracts, integration testing, end-to-end validation, deployment, code reviews, and release readiness.
- Coordination rule: Agree on API schemas, request/response formats, error codes, and authentication requirements before implementing dependent features. Use feature branches and small, frequent merges.

Important: Since this is a clinical system, the backend developer must retain responsibility for server-side authorization, audit logging, transactional consistency, and clinical decision rules. The frontend may display clinical warnings but must never be the sole enforcement point for safety-critical rules.

6. Known Rabbit Holes & Time-Boxes

The following high-risk items are explicitly time-boxed to prevent schedule slippage:

1. **Payment Gateway SDK Integration (Time-Boxed to 0 Hours):**
   - *Rabbit Hole:* Implementing Razorpay/Stripe webhooks, signature verification, and reconciliation edge cases can consume 1–2 full weeks.
   - *Boundary:* **Zero payment gateway SDK code in Cycle 1.** Multi-point billing supports manual recording of Cash, UPI UTR numbers, Card authorization codes, and Cheques only.
2. **Physical Lab Analyzer Hardware Drivers (Time-Boxed to 0 Hours):**
   - *Rabbit Hole:* Parsing proprietary RS-232 / ASTM / HL7 serial protocols for haematology analyzers requires physical laboratory machine access and custom driver development.
   - *Boundary:* **Technician manual parameter entry only.** Standard multi-parameter form with automated min/max/panic range validation.
3. **External LLM Prompt Engineering & Ambient AI (Time-Boxed to 0 Hours):**

- *Rabbit Hole:* Fine-tuning OpenAI/Claude prompts or Whisper STT audio pipelines introduces latency delays, hallucination risks, dependency on cloud internet, and non-deterministic medical outputs.
- *Boundary:* **Deterministic relational SQL rules and heuristic scoring only** (`cdss_ddi_rules`, `cdss_vital_rules`). Sub-5ms execution, zero hallucination, 100% auditable, and zero external API costs.

4. **Complex Supply Chain & Vendor PO Lifecycles (Time-Boxed to 0 Hours):**
   - *Rabbit Hole:* Vendor purchase orders, goods receipt notes (GRN), and multi-warehouse transfers.
   - *Boundary:* **Simple batch entry with FIFO deduction.** Pharmacist logs incoming lots (Batch#, expiry, quantity) and dispenses directly.

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7. Explicit No-Goes for Cycle 1

The following features are **explicitly excluded** from Cycle 1:

- ❌ **Ambient AI Scribing & Audio Speech-to-Text:** No Whisper, audio streaming, microphone recording, or ambient room listening in this MVP.
- ❌ **Non-Deterministic Large Language Models (LLMs):** Zero OpenAI, Claude, or generative AI prompt calls. All clinical safety and summaries are 100% deterministic rules and code heuristics.
- ❌ **Telemedicine & Video Consultation:** In-person outpatient and inpatient clinical workflows only.
- ❌ **Native Biometric Fingerprint Scanners:** Standard keyboard/mobile demographic entry only.
- ❌ **Patient-Facing Mobile App / Web Portal:** Staff-facing clinical and administrative applications only.
- ❌ **Multi-Hospital Tenant Switcher UI:** Schema is forward-compatible with `hospital_id`, but the user interface is scoped strictly to a single hospital facility in Cycle 1.
- ❌ **Direct WhatsApp / SMS Gateway Messaging:** Digital slips and reports are downloaded or printed directly from the workstation.

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8. Cycle 1 Compliance & Release Gate

At the end of Week 4, the software must pass the following release gates before final deployment to the hospital:

1. **Safety Gate:** 100% of CDSS DDI rules verified against test drug pairs (Warfarin + Aspirin must reliably block without justification).
2. **Financial Gate:** IPD consolidated discharge invoices must balance to the exact rupee (`net_amount == subtotal - discount + tax == paid + balance`) across all 10 charge heads.
3. **Privacy Gate:** Zero patient names or diagnoses present in the waiting room display SSE stream.

4. **Reliability Gate:** Local edge Docker fallback successfully boots and accepts emergency check-ins while the Contabo connection is severed.